A Detailed Analysis of Film Cooling Physics: Part III—Streamwise Injection With Shaped Holes

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The physics of the film cooling process for shaped, streamwise-injected, inclined jets is studied for blowing ratio (M = 1.25, 1.88), density ratio (DR = 1.6), and length-to-diameter ratio (L/D = 4) parameters typical of gas turbine operations. A previously documented computational methodology is applied for the study of five distinct film cooling configurations: (1) cylindrical film hole (reference case); (2) forward-diffused film hole; (3) laterally diffused film hole; (4) inlet shaped film hole, and (5) cusp-shaped film hole. The effect of various film hole geometries on both flow and thermal field characteristics is isolated, and the dominant mechanisms responsible for differences in these characteristics are documented. Special consideration is given to explaining crucial flow mechanisms from a vorticity point of view. It is found that vorticity analysis of the flow exiting the film hole can aid substantially in explaining the flow behavior downstream of the film hole. Results indicate that changes in the film hole shape can significantly alter the distribution of the exit-plane variables, therefore strongly affecting the downstream behavior of the film. Computational solutions of the steady, Reynolds-averaged Navier–Stokes equations are obtained using an unstructured/adaptive, fully implicit, pressure-correction solver. Turbulence closure is obtained via the high-Reynolds-number k–ϵ model with generalized wall functions. Detailed field results as well as surface phenomena involving adiabatic film effectiveness (η) and heat transfer coefficient (h) are presented. When possible, computational results are validated against corresponding experimental cases from data found in the open literature. Detailed comparisons are made between surface and field results of the film hole shapes investigated in this work; design criteria for optimizing downstream heat transfer characteristics are then suggested.

1 Introduction

An understanding of jet-in-crossflow interaction is crucial in the design of hot section components in modern gas turbine engines. The temperature of the gases entering the turbine section is typically near the melting point of the alloys used in the turbine airfoils and endwalls; to prevent these components from failing at such elevated temperatures, they are commonly film cooled to isolate the metal from the hot gases. In addition, the geometry of film cooling holes is often modified in some manner to improve cooling performance, thus allowing for better metal protection and/or a decreased coolant supply; however, these configurations are often implemented with weak understanding of flow and heat transfer characteristics in and near the film hole. This study focuses on inclined discrete jet film cooling physics and the effect of film-hole shaping on both flow and thermal field characteristics. Detailed field data, the adiabatic effectiveness, and the surface heat transfer coefficient will be used to evaluate the advantages and disadvantages of film-hole shaping. Emphasis is placed on determining, with the help of vorticity analysis, the dominant mechanisms responsible for the performance of the various film hole configurations. Specifically, the present investigation aims to examine several film-hole shapes, pinpoint the flow mechanisms responsible for determining downstream heat transfer, and suggest ways of using film hole geometry to exploit or attenuate these flow mechanisms.

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2 Literature Review

The interaction of jets in crossflow has been heavily researched both experimentally and computationally. Some relevant references applicable to cylindrical film holes are discussed in detail in Part I of this paper; only references specific to shaped film holes are discussed here.

Goldstein et al., (1974) were among the first to pioneer the use of shaped film holes for improved film cooling performance. The performance of inclined holes with a 10 deg laterally flared exit was compared with the performance of streamwise inclined cylindrical film holes, both with an L/D of 5.2 and P/D of 3 and 6. Flow visualization studies showed that with and without the mainstream flow, the shaped hole provided much better lateral coverage of the test surface and also attenuated jet liftoff. Effectiveness data confirmed the observations that the shaped film hole afforded better lateral coverage and better centerline effectiveness. In addition, the shaped film holes provided significant improvements in cooling performance at any M and DR. The authors hypothesized that the increased exit area of the shaped holes was responsible for slowing the coolant flow such that less penetration through the oncoming boundary layer and into the mainstream occurred.

Papell (1984) experimentally studied a novel shape for use in film cooling. Papell found that shaping the hole with a cusp (similar in appearance to a kidney bean) induced strong longitudinal vortex structures within the film hole; he hypothesized, with the aid of flow visualizations, that creating these vortical structures enabled the crossflow to use its energy to force the jet down to the surface, rather than itself creating the counterrotating vortices. Further, he postulated that the placement of the cusp on the leeward side (TE) forced the film hole secondary flow to rotate in a direction opposite of that traditionally observed in cylindrical

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film holes. Papell offered evidence in the form of adiabatic effectiveness and film coverage data to support his findings that his cusp-shaped holes provided better film cooling performance.

Makki and Jakubowski (1986) presented downstream heat transfer results for a film hole with a trapezoidal shaped expansion. Using the downstream heat transfer coefficient versus the heat transfer coefficient of an undisturbed boundary layer, they showed that the shaped film hole consistently provided better heat transfer characteristics than simple cylindrical holes with the same metering section. Makki and Jakubowski reported that the shaped holes offered up to 23 percent better film cooling performance than the corresponding cylindrical hole. Much guesswork was applied in determining the physical mechanisms responsible for this improved film cooling performance; like Goldstein et al. (1974), the authors attributed much of the improved film cooling performance to the decreased momentum flux at the jet exit due to the diffusing shape of the trapezoidal film hole.

Schmidt et al. (1996) and Sen et al. (1996) presented two companion papers in which the effect of adding a 15 deg forward diffusing exit to a streamwise oriented film hole was investigated. Moderate M and DR typical of gas turbines were used, along with a realistic L/D of 4.0 and P/D of 3.0. They found that the exit diffused film hole demonstrated better spread of adiabatic effectiveness than the cylindrical counterpart; however, by $x/D = 15$ the expanded and round compound angle holes had essentially the same level of effectiveness. From the heat transfer coefficient standpoint, the forward-expanded hole performed poorly, presumably because of the increased interaction between the jet and mainstream. A key conclusion shown in this work was that to determine overall film cooling performance, it is crucial to obtain knowledge of both the adiabatic effectiveness and the heat transfer coefficient.

Thole et al. (1998) studied the effect of film hole geometry on the film cooling flowfield. The film hole shapes examined in this work were a cylindrical (reference) case, a hole with a laterally expanded exit, and a hole with a forward–laterally expanded exit. M and DR were unity, and external mainstream Mach number and internal coolant supply Mach number were nearly matched at approximately 0.3. Results from this study showed that by expanding the exit of the film holes, the penetration of the jet as well as the severe shear regions in the near field were significantly reduced when compared to the reference cylindrical hole. Turbulence levels, although similar in magnitude between the three cases, were found to originate from different sources; therefore, the flow characteristics were very dependent on the film hole geometry.

Haven and Kurosaka (1996) documented the effects of film hole shape on the near-field vortical structures characteristic of a jet-in-crossflow interaction. Note that the study concentrated solely on fluid mechanics (no heat transfer), and the shaped jets were injected normal to the test surface. The shapes investigated were simple two-dimensional shapes without a metering section and with equal cross-sectional area; the shapes were: (1) circular, (2) square, (3) rectangular, and (4) elliptical. Limited quantitative data were taken; rather, flow visualizations were conducted for various film hole shapes to determine the effect on downstream vortical structures. The authors proposed that the counterrotating vortices downstream of the jet exit plane had a significant influence on jet liftoff, by the mutual induction lift generated by these structures. They concluded that hole geometries that cause these vortices to form farther apart tend to inhibit jet detachment, since the lift generated by mutual induction is less. One of the principal mechanisms affecting the formation of the counterrotating vortices was the vorticity contained in the film hole sidewall boundary layers. This vorticity will be discussed in detail in the present paper.

Hyams et al. (1996) studied the effects of slot jet shaping on the heat transfer downstream of a slot jet. It was found that shaping of the slot inlet and exit provided significant gains in the film cooling performance. Flow mechanisms responsible for this behavior, such as the exit plane momentum characteristics and sources of turbulence within the film hole, were documented in detail. Both adiabatic effectiveness and heat transfer coefficient data were obtained in this study.

The effect of film hole shaping and orientation for transonic conditions was investigated by Wittig et al. (1996). Measurements and computations were performed for a cylindrical film hole, laterally expanded film hole, and a forward–laterally expanded film hole. Effectiveness data and flowfield data were only presented for the cylindrical reference case, however. Velocity field computations with a multiblock structured grid provided fair agreement with the measured results along the centerplane outside of the film hole; however, predictions within the film hole and at the film hole exit plane were (according to the authors) poor. Only one experimental visualization was presented for any shaped case; the bulk of the results discussed only the cylindrical film hole.

No detailed computational studies of three-dimensional shaped film holes exist in the open literature at this time. In addition, no

Nomenclature

AR = area ratio = $A_{\text{metering}}/A_{\text{exit-plane}}$
CUSP = cusp shaped film-hole
D = diameter of film-hole
DR = density ratio = ρ_j/ρ_∞
FDIFF = film hole with a forward diffused exit
h = heat transfer coefficient = $q''/(T_w - T_{aw})$, W/m^{2}K
ISHAP = film hole with inlet shaping
L/D = length-to-diameter ratio of film hole
LDIFF = film hole with laterally diffused exit
LE = leading edge of film-hole exit plane
M = blowing (or mass flux) ratio = $(\rho V)_j/(\rho V)_\infty$
Pr = Prandtl number
P/D = pitch-to-diameter ratio of film hole
q'' = surface heat flux per unit area, W/m^2
Re = Reynolds number
REF = cylindrical reference film-hole case
St = Stanton number
T = local fluid temperature
TE = trailing edge of film-hole exit plane
tke = tubulence kinetic energy
TL = turbulence level (in percent) = $(\frac{2}{3}*\text{tke})^{1/2}/V_\infty*100$
V = local fluid velocity magnitude = $|\vec{V}|$
$\vec{V}$ = local velocity vector = $u\hat{\imath} + v\hat{\jmath} + w\hat{k}$
VR = velocity ratio = V/V_∞
x = coordinate in the streamwise direction
x_0 = unheated starting length
y = coordinate normal to the test surface
y^+ = nondimensional wall distance = $y\sqrt{\tau_w/\rho_w}/\nu_w$
z = coordinate in the lateral direction
α = coolant jet injection angle
δ = boundary layer thickness
η = adiabatic effectiveness = $(T_\infty - T_{aw})/(T_\infty - T_j)$
ν = kinematic viscosity
ρ = fluid density
$\vec{\xi}$ = total vorticity vector = $\xi_x\hat{\imath} + \xi_y\hat{\jmath} + \xi_z\hat{k}$

Subscripts

∞ = mainstream conditions at crossflow inlet plane
aw = adiabatic wall
fh = condition inside film hole
j = nominal jet conditions
w = conditions at wall

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systematic investigation has been conducted concerning different film hole configurations and their effect on the complex three-dimensional flow mechanisms present both inside of the film hole and in the near-field interaction.

3 Present Contributions

The primary objective of this work is to examine thoroughly the physics of a row of inclined, discrete, shaped jets in crossflow, while pinpointing the physical flowfield and heat transfer differences between a cylindrical reference case and selected shaped film holes. These three-dimensional studies include several shapes as suggested by the gas turbine industry and as chosen at the discretion of the principal investigators. Objectives of this study are to:

- determine the effect of several industry-relevant film hole shapes on downstream heat transfer;
- investigate and document the physical mechanisms responsible for changes in film cooling performance for shaped film holes, so that better designs can be achieved;
- present both adiabatic effectiveness and heat transfer coefficient downstream of shaped-hole film cooling configurations;
- demonstrate that CFD can be used to capture flow and heat transfer details accurately and dependably;
- validate a consistently accurate computational methodology and transfer this technology to the gas turbine industry;
- establish criteria for more effective design for shaped film holes.

4 Computational Methodology

The computational methodology used in this study is identical to the methodology documented in Walters and Leylek (1997) and will not be repeated here. The computational hierarchy used encompasses a proper film cooling model, an exact geometry, a high-quality, high-density computational mesh, high-order discretization schemes, and turbulence modeling that is suitable for the problem at hand. The simulations are carried out in a fully elliptic manner using an implicit, pressure-correction based, unstructured/adaptive grid Navier–Stokes solver.

5 Experimental Details

Data from the University of Texas (Schmidt et al., 1996) are used for the cylindrical and forward-diffused film holes. Schmidt et al. (1996) document adiabatic effectiveness results downstream of streamwise-injected cylindrical and forward-diffused film holes, and the experimental apparatus is detailed in that work. The film holes used in this study are injected at $\alpha = 35$ deg, and spaced at $P/D = 3$. The exact geometry of these film holes is reproduced in the computations and is shown in Fig. 1. The density ratio of these interactions was $DR = 1.6$; blowing ratios tested were $M = 1.25$ and $M = 1.88$.

6 Simulation Details

6.1 Definition of Shaped Film Holes. Five basic film hole shapes are investigated in this work. The configurations are: (1) cylindrical (REF), (2) forward diffused (FDIFF), (3) laterally diffused (LDIFF), (4) inlet shaped (ISHAP), and (5) cusp shaped (CUSP). Schematics of each of these shapes is shown in Fig. 1.

Each film hole geometry (except for the CUSP case) is simulated for two blowing ratios, and adiabatic effectiveness and heat transfer results are obtained for each. Table 1 below shows an overview of the simulations performed in the present work. Boundary condition specifics are discussed in the following section.

6.2 Modeling and Boundary Conditions. For each of the film hole shapes described in this work, the boundary conditions

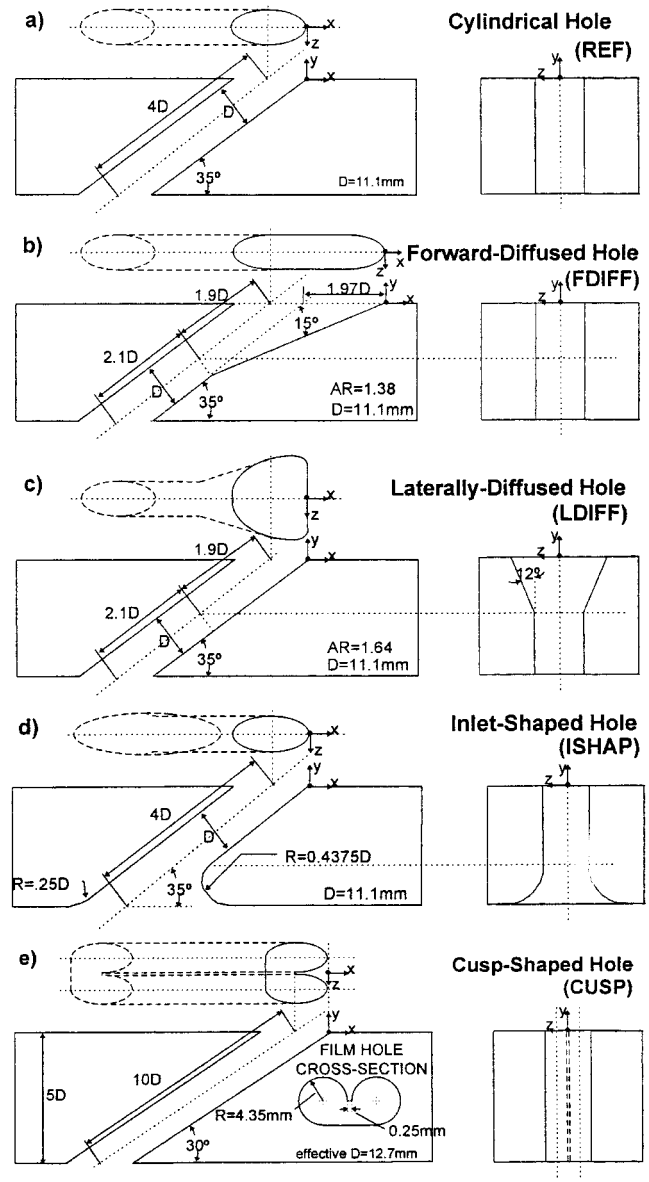

Fig. 1 Schematics of the selected film hole shapes show the geometry of each configuration in three orthogonal planes

are identical except for differences noted in the following paragraphs; a schematic of the domain and boundary conditions is shown in Fig. 2. Symmetry conditions are imposed at the longitudinal centerline of the jet, as well as at $P/D = 1.5$. These boundary conditions enforce streamwise injection of the jet and a lateral spacing of $P/D = 3$ for an infinite row of jets. Due to the height of the wind tunnel used in the experimental counterparts, a zero-normal-gradient (slip-wall) condition is applied at the top of

Table 1 Overview of the simulations performed in the present work

Film Hole Shape	M	DR	test surface
Cylindrical (REF)	1.25, 1.88	1.6	adiabatic, const. q"
Forward Diffused (FDIFF)	1.25, 1.88	1.6	adiabatic, const. q"
Laterally Diffused (LDIFF)	1.25, 1.88	1.6	adiabatic, const. q"
Inlet Shaped (ISHAP)	1.25, 1.88	1.6	adiabatic, const. q"
Cusp Shaped (CUSP)	1.25	1.08	adiabatic, const. q"

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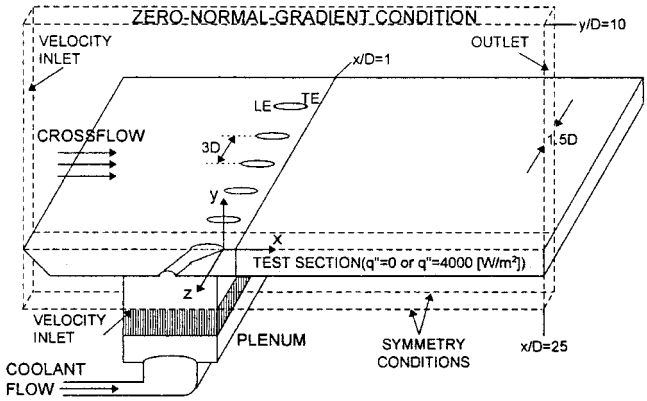

Fig. 2 A sketch of the computational domain shows the boundary condition scheme used for each shaped case

the domain. This zero-gradient wall is imposed at $y/D = 10$ for all simulations to reduce the size of the computational domain; it is found that all gradients in the y direction are negligible far before this limit is reached.

An outlet (convection) boundary condition is imposed at $x/D = 25$. The plenum inlet velocity (coolant air) is set to achieve $\mathrm{Re}_{fh} = 18{,}700$ for every case; this matches the velocity range used in Schmidt et al. (1996), and the exact velocities are documented in the following paragraphs. Various M are attained by varying the crossflow (air) inlet flux, as described below. TL at the plenum inlet is assumed to be 0.1 percent. For all cases, the length scale for each inlet was taken to be one-tenth of the inlet extent. Following the Schmidt et al. (1996) cryogenic setup, the crossflow inlet temperature is specified as 300 K, and the plenum coolant temperature is set at 187.5 K; these temperatures force $DR = 1.6$ for all cases. All walls are adiabatic, except for the test section in the active wall (heat transfer coefficient) simulations; in these cases, the wall heat flux is set to 4000 W/m^2. The heated test section starts at $x/D = 1$ for all cases.

The geometry for the REF and FDIFF cases is taken from the work of Schmidt et al. (1996); therefore, the plenum geometry and the upstream section match the experimental setup described there. This plenum is 50.8 mm high and 101.6 mm wide, and the inlet velocity is set to 0.335 m/s. A uniform velocity inlet condition is imposed at the beginning of the 237 mm upstream section, and a turbulent boundary layer is allowed to develop from this point. Turbulence intensity at the crossflow inlet is assumed to be 0.2 percent, as per the documentation of Schmidt et al. (1996).

For the LDIFF and ISHAP cases, the plenum geometry matches the geometry used in in-house experiments; this geometry is documented in Farmer et al. (1997). The plenum velocity is set to 0.2855 m/s. The upstream section is extended $10D$ prior to the upstream edge of the film holes, in accordance to the location at which velocity and turbulence intensity profiles were experimentally acquired. Experimentally measured profiles for velocity and turbulence intensity are imposed at this location. The experimentally measured $\delta/D = 0.55$ condition at the film hole is matched.

The plenum used in the Schmidt et al. (1996) experiments is also used in the CUSP case, since the actual geometry of the plenum was not specified in Papell (1984). The crossflow inlet velocity profile is constructed to match $\delta/D = 1.3$ at the film hole breakout and the inlet TL is set to 3 percent, as documented in Papell's study. Temperature at the crossflow inlet is set to 295 K, and the coolant is injected at 273 K; these temperatures set $DR = 1.08$. Since D, L/D, Re_{fh}, and DR are different from the other simulations in this work, no cross comparisons involving the CUSP case and other shaped cases are possible.

Most notable of the boundary condition and modeling differences in these simulations are the two different plenum geometries used. However, since the flow in the supply plenum is largely stagnant, differences in plenum geometry are very unlikely to affect film hole inlet-plane conditions.

6.3 Convergence. A fully implicit, pressure-correction, unstructured/adaptive grid code was used to perform all processing of the computational simulations. Figure 3 shows the ISHAP surface grid used in this study; this grid shows the typical cell quality and density of the mesh for all the computations. Preprocessing and post-processing were performed on Sun and SGI workstations, while computations were performed on a 64 CPU Intel Paragon supercomputer. Simulations were typically run using 32–56 CPUs, which required about 6 hours per 300 iterations; approximately 1500 iterations were necessary to reach a fully converged state. Convergence was established when the mass and energy imbalance in the entire computational domain was less than 0.01 percent of inlet fluxes, normalized residuals had fallen at least three orders of magnitude, and no change was observed in any field result. Grid independence was assessed by repeatedly refining the grid (based on field gradients) until no observable changes were apparent in the field results. Typical background grids started at 250,000 cells; fully converged and grid independent results were typically 400,000 cells.

7 Results

7.1 Vorticity Analysis of Film Cooling Interactions. Part I of this four-part series highlights that the most dominant mechanism in determining film cooling performance is the longitudinal vortex structure formed aft of the jet injection site. Further, it is stated that boundary layer vorticity from the film hole controls the strength of these longitudinal vortices. The physical explanations of the important flow mechanisms in the following sections rely heavily on vorticity analysis; consequently, it is necessary to define the concepts that are used to pinpoint the formation of film hole boundary layer vorticity.

Note that in all discussions of vorticity that follow, only the positive z side of the film hole is considered. Due to symmetry, the same mechanisms (but with opposite sign) occur on the negative z direction side.

For a streamwise-injected film hole, the contributors to vorticity aligned in the streamwise direction (hereafter referred to as "aligned vorticity" and with the symbol ξ_x) are illustrated in Fig. 4(a). Because of secondary motion within the film hole, the interior boundary layer travels in a helical motion. The resultant vorticity vector, shown as the dotted line in Fig. 4(a), can therefore be decomposed into a component due to throughflow that is perpendicular to the flow direction, and a component due to secondary flow that is in the direction opposite to the flow direction. The resultant of these two components is then resolved to a coordinate system aligned with the predominant crossflow direc-

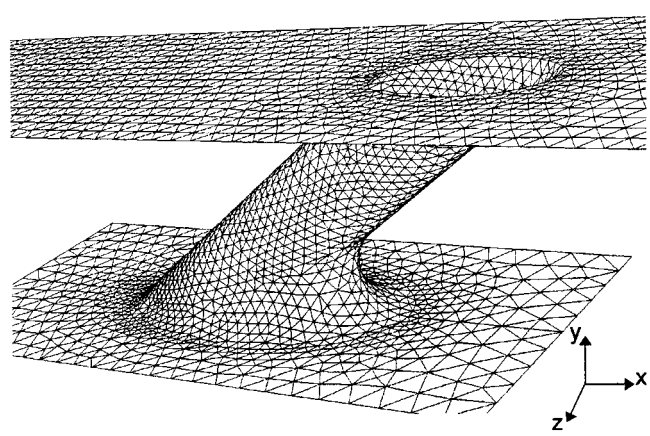

Fig. 3 A photo of the ISHAP computational grid shows typical grid quality and resolution

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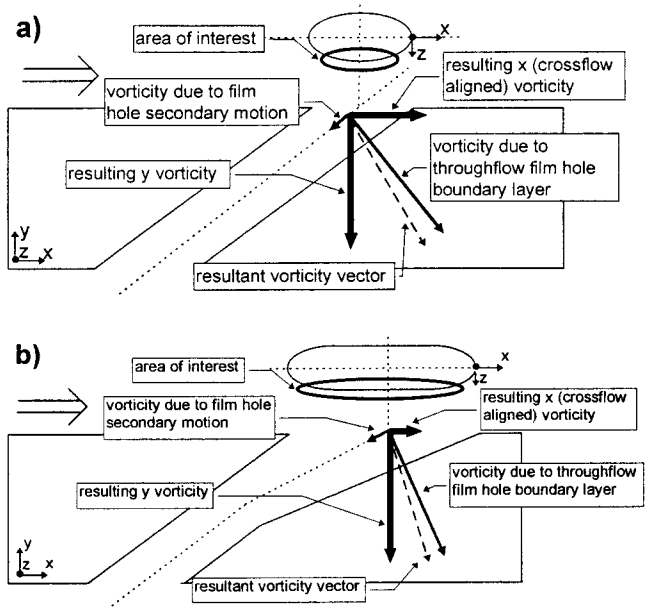

Fig. 4 Sketch of the film hole boundary layer vorticity shows the manipulation of streamwise aligned vorticity by film-hole geometry

tion to determine the aligned vorticity. This aligned vorticity is denoted by ξ_x. Inspection of Fig. 4(*a*) illustrates two ways to decrease the aligned vorticity: (1) decrease the throughflow vorticity, or (2) increase the secondary vorticity.

Figure 4(*b*) illustrates the effect on the aligned vorticity as the exiting flow angle decreases. The throughflow vorticity is tilted downward, where it cannot contribute a great deal to the aligned vorticity. By the same argument, it is obvious that increasing the flow angle would not only increase the *y* momentum and penetration into the crossflow, but would also tilt the resultant sidewall vorticity vector forward; this would (detrimentally) increase the magnitude of crossflow-aligned vorticity.

7.2 Cylindrical Film Hole (REF). To form adequate comparisons and isolate mechanisms in the upcoming shaped film-hole cases, a solid well-defined reference case must be formulated. A separate reference case (an inclined cylindrical film hole) is simulated in this study to provide proper basis for comparison. The findings reported in this section are largely a summary of the work documented in Part I. Obviously, only the most relevant items and/or conclusions in this reference case are summarized here; note that the results of these separate simulations substantiate the findings of Part I of the present work. Following the reference case, separate sections identify the key mechanisms dominating the downstream heat transfer for the shaped film holes investigated in this paper.

Key conclusions from Part I include the effect of film-hole boundary layer vorticity on the downstream development of the inclined jet. The concept of explaining near-field interactions in terms of film hole boundary layer vorticity finds critical application in the area of shaped film holes. Reducing the strength of crossflow-aligned vorticity (ξ_x) exiting the film hole, and therefore the resulting counterrotating secondary structure, can prevent the liftoff of the coolant/wake structure downstream.

Figure 5 shows the computed versus measured η for the REF case at $M = 1.25$ and $M = 1.88$. Overall, the agreement between predicted and measured effectiveness is good. This lends credibility to the quality of simulations presented in the present paper. In the near field ($x/D < 3$), η is overpredicted somewhat, presumably because of a hypothesized recirculation region immediately downstream of the film-hole exit; the use of wall functions in the simulations do not allow the grid resolution necessary to capture such a phenomenon. As the jet moves downstream, the computations seem to slightly underpredict the diffusion rate of the lifted-off jet down to the test surface.

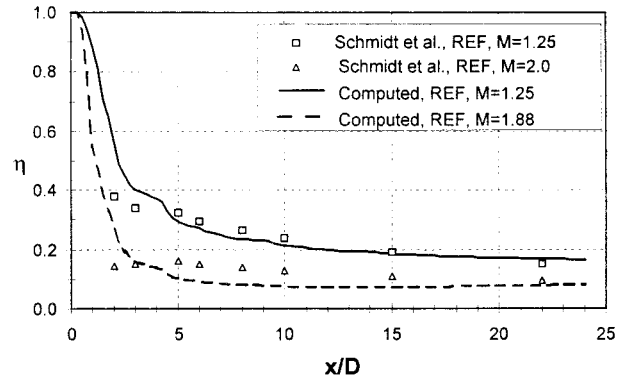

Fig. 5 A validation plot of centerline effectiveness shows good agreement between computational and experimental data

7.3 Forward-Diffused Film Hole (FDIFF). At the inlet of the FDIFF film hole, the same mechanisms are seen to exist as present in the REF case. The film-hole geometry is identical to the REF case within the metering section until the diffusing section is reached; therefore, the severe flow turning, the low-momentum zone, and jetting region are apparent as in the REF case. In fact, the consistency of the computational simulations is excellent in this area; *exactly* the same magnitude of all flow quantities for the REF and FDIFF cases are computed for the inlet and metering section of the film hole. This observation implies that the inlet development is independent of any elliptic effect by the crossflow impingement on the coolant jet, and of any effect by the diffusing section.

As the coolant flow enters the diffusing section within the FDIFF film hole, a small separation is visible on the downstream wall of the film hole; also, the entire coolant flow slows drastically as the flow area is widened. Because of the use of wall functions in these simulations, the overall extent of the separation may not be exact, and may contribute more to turbulence generation within the film hole (because of shear layers over the separation zone) than what is computed. However, it is not expected that this separation will significantly alter the distribution of the exit plane variables.

Because of the wide diffusing section, the effective *VR* of the FDIFF case is drastically lower than that of the corresponding REF case, as shown in Fig. 6(*b*). Note that the nominal $VR = 0.78$ for the $M = 1.25$ simulations. The higher-momentum jetting region is still visible on the windward side (LE) of the film hole (by examination of *VR* and α in Fig. 6(*b*) and Fig. 7(*b*), and low-momentum fluid exits from the leeward side (TE) of the film hole.

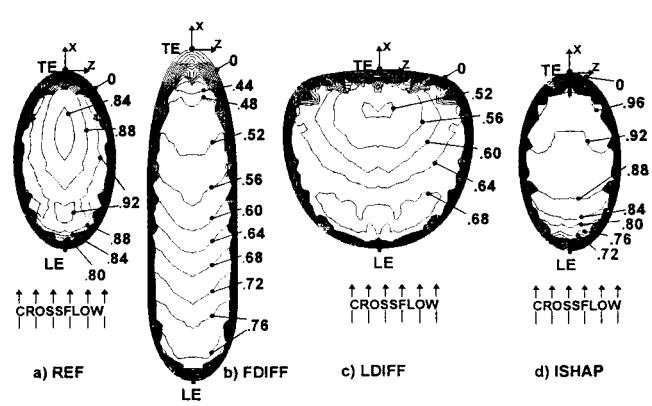

Fig. 6 A comparison of the exit plane *VR* distribution shows the low-momentum content of the diffused holes

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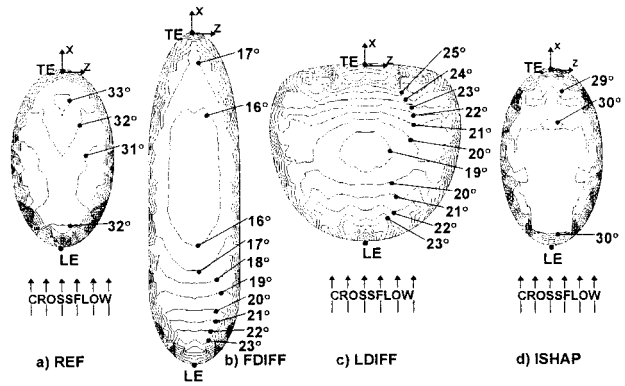

Fig. 7 Contours of the flow exit angle (α) at the exit plane show the various effective injection angles of each film hole shape

Also, as shown in Fig. 7(*b*), the flow exit angle is typically one-half that of the reference cylindrical hole; these two mechanisms contribute to an exceptionally low exit-plane *y*-momentum content of the FDIFF film hole. Although total *y*-momentum content is lower, the jetting region (comparable in magnitude to the REF case) controls the trajectory of the coolant jet. As shown in Fig. 12, the location of the jet-film interface is similar to the REF case. Note that, due to the definition of the origin in the FDIFF case (see Fig. 1(*b*)), the shear layer above the film has approximately $2D$ greater distance in which to develop. So, the jet trajectory is actually slightly below the trajectory predicted for the REF case.

Notably absent from the FDIFF case is the coolant wake observed and documented in Part I. A strong low-pressure region downstream of the TE of the film hole exit plane is absent; therefore, coolant from the film hole boundary layer is injected uniformly into the mainstream boundary layer as shown in Fig. 8(*b*). This behavior is in stark contrast to the REF case, where coolant fluid in the boundary layer is drawn toward the centerline, is slowed further by an adverse pressure gradient, and coalesces into a coolant wake that is subsequently lifted by the action of the counterrotating vortices that appear downstream of the injection site. The inward bending and lifting of the film-hole boundary layer fluid is strictly due to the action of the weak longitudinal vortices seen in the FDIFF case. The absence of a strong exit-plane pressure gradient and subsequent lack of a pronounced coolant wake is directly related to the low *y*-momentum content of the coolant exiting the FDIFF film hole.

The ξ_x levels contained within the exiting boundary layers on the lateral sides of the FDIFF film hole is much lower than the REF counterpart, as shown in Fig. 10(*b*). This difference in vorticity characteristics is due to: (1) velocity gradients along the side walls are relaxed due to decreased throughflow film-hole velocities; and (2) the shallow exit angle, especially along the side walls, afforded by the forward diffusing section. Each of these two mechanisms contributes greatly to the reduction in streamwise vorticity seen in the FDIFF case. More detail on this topic is discussed in section 8.2. Note that the length of the film-hole sidewall tends to add more ξ_x to counteract the advantages enumerated above; however, this effect, along with the constant dissipation of vorticity near the wall, is not pronounced enough to add significantly to the *x*-direction vorticity from the film hole. As a result of the much decreased ξ_x ejected and coalescing at the sides of the film hole, the strength of the downstream secondary motion (the typical counterrotating vortex pair) is decreased as shown in Fig. 11(*b*). Eliminating most of this secondary motion reduces unwanted induction lift and allows the coolant jet to retain proximity to the test surface, and also prevents the convection of entrained hot crossflow fluid underneath the coolant jet. A detailed discussion of this important phenomenon is given in section 8.2 of this paper and also in the companion paper designated as Part I.

The centerline temperature profiles in Fig. 9 show that the coolant core remains close to the wall, in contrast to the lifting apparent in the REF case. Downstream of the jet–crossflow interaction, the dominant mechanism becomes the generation of turbulence located throughout the film. Profiles of *TL* in Fig. 13 demonstrate the large difference in far-field turbulence production between the REF and FDIFF cases; this difference in behavior arises because of the large mismatch in exit velocity created by the FDIFF case; the crossflow attempts to "drag" the low-momentum jet along as it travels downstream. In the near field, *TL* is very similar; however, as the coolant film moves downstream, significant amounts of turbulence are generated throughout the film in the FDIFF case (an indication of the vertical velocity gradient can be assessed easily by the ξ_z profiles in Fig. 12). This large length-scale turbulence contributes a great deal to the diffusion rate of the crossflow energy into and across the coolant jet. Hence, the coolant is dissipated downstream not only by strong mixing at the film–crossflow interface, but also by mixing throughout the film itself.

As shown in Fig. 17, the centerline performance of the FDIFF film hole is excellent compared to the REF case; however, little spreading (and so small test plate coverage) of the temperature core exists. The underprediction of film spreading rate due to the anisotropic nature of turbulence downstream of a jet injection is a common feature of many CFD simulations; this issue is discussed in Part I. The heat transfer coefficient for the FDIFF case is consistently lower (at all x/D) than the REF case for $M = 1.25$ and $M = 1.88$.

7.4 Laterally Diffused Film Hole (LDIFF). Like the FDIFF case, the LDIFF case has identical film hole inlet-plane conditions as the reference case. The first effects of the film hole shaping are felt very close to the laterally diffusing section. No separation is seen on the sides of the 12 deg diffusing section; the flow smoothly negotiates the film hole area change and begins to slow as the area is widened. Note that Thole et al. (1998) observed high turbulence levels at the lateral sides of the film hole exit plane, which would imply separation within the 14 deg laterally diffused film hole; however, Re_{fh} in their experiments ($\mathrm{Re}_{fh} \approx 52{,}000$) was three times higher than these simulations ($\mathrm{Re}_{fh} \approx$

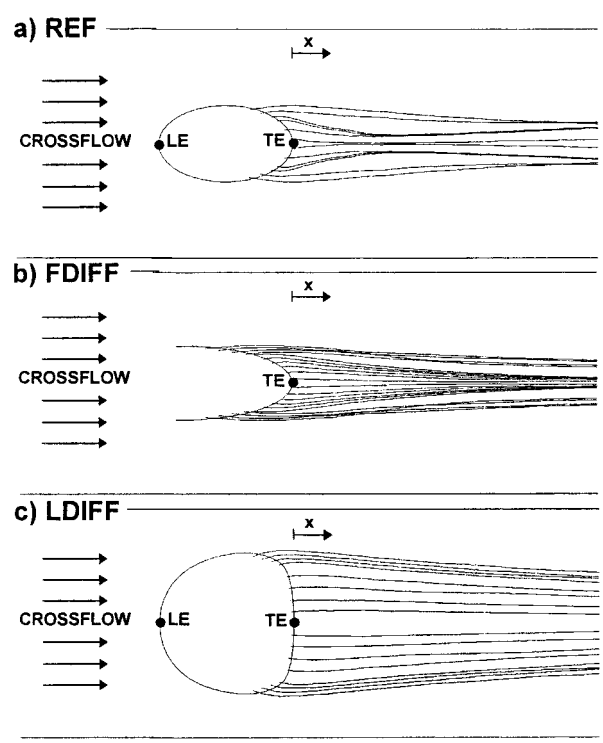

Fig. 8 Pathlines from the film-hole boundary layer show the presence or absence of a coolant wake for the REF, FDIFF, and LDIFF cases

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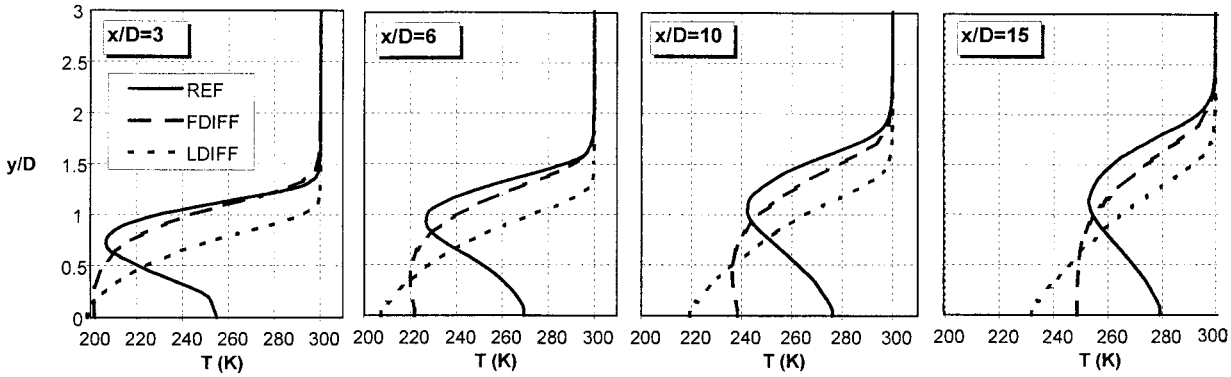

Fig. 9 Centerline temperature profiles at $x/D = 3$, $x/D = 6$, $x/D = 10$, and $x/D = 15$ for the REF, FDIFF, and LDIFF cases show the overall coolant characteristics

18,700). In addition, coolant was introduced into the film hole from a coflowing channel, which would largely eliminate the effect of the inlet separation/low-momentum zone on film hole flow characteristics. Like the FDIFF film hole, the effective *VR* at the exit plane is much smaller (Fig. 6(*c*)). The distribution of α is complex as shown in Fig. 7(*c*), but shallower than the REF case as a whole. The smallest α, near the center of the film hole, is associated with low-momentum fluid aft of the inlet separation zone; higher α is associated with the jetting region at the upstream edge and the metal angle on the downstream edge.

At the exit plane, a smoother and less severe pressure gradient exists. This again highlights that the crossflow sees a much less severe flow blockage than the REF case. Note that this condition occurs even though the LDIFF film hole presents a wider lateral area to the crossflow; the trajectory of the jet is very low, so the coolant is effectively injected into the lower-momentum crossflow boundary layer. As a result, film hole boundary layer fluid is not drawn to the centerline to form a coolant wake; rather, it is injected along the test section in the streamwise direction, as shown in Fig. 8(*c*). Notably, the coolant wake observed in Part I is not apparent in the LDIFF case; the fluid that makes up this wake is spread evenly over the test surface.

An important feature of the LDIFF film hole, as shown in Fig. 10(*c*), is the reduction afforded in streamwise-aligned vorticity. Because of the decreased velocity gradients along the walls of the film hole, the vorticity contained in the film hole boundary layer is much less in magnitude; it is also physically separated in the *z*-direction as it is deposited on the test plate. The flow angle also significantly affects the magnitude of ξ_x, as in the FDIFF case; as shown in Fig. 7(*c*), the flow direction exiting the LDIFF film hole is much less than the REF case. Obviously, the lower *x*-direction vorticity content causes the longitudinal vortices to be weakened in strength (Fig. 11(*c*)). Weak longitudinal vortices allow the film to remain firmly attached to the surface, and the film has a greater chance to spread along the surface (rather than being "pinched" as documented in Part I). In addition, since the vortices are physically separated, less induction lift (which lifts the coolant core) results from their presence. Profiles of *z* vorticity in Fig. 12 show the difference in liftoff characteristics for the REF and LDIFF cases. Positive *z* vorticities denote liftoff and the infiltration of high-momentum, high-temperature crossflow fluid underneath the coolant jet. In contrast to the coolant liftoff apparent in the REF case, the LDIFF case demonstrates lateral vorticity characteristics similar to a boundary layer, which is ideal for a film cooling application.

As the coolant moves downstream in a wakelike manner in the far-field, the behavior is much like the FDIFF case. Velocity gradients in the shear layer above the jet and throughout the film fuel turbulence in that area, augmenting the diffusion of crossflow energy into the coolant layer. This effect is much smaller in the $M = 1.88$ case, because the effective *VR* of the jet and crossflow is more closely matched.

The laterally averaged η, shown in Fig. 18, is approximately three times that of the REF case; this drastic improvement is attributed to (1) an excellent lateral coverage over the test plate as seen in Fig. 16(*c*), (2) shallow injection that is kept largely into the low-momentum crossflow boundary layer, and (3) much weaker, laterally separated longitudinal vortices to prevent crossflow entrainment and downstream coolant liftoff. Note that the slope of the LDIFF curve in Fig. 18 decreases throughout the length of the domain; this implies that the coolant is slowly dissipated by the diffusion mechanisms described in the previous paragraph. In

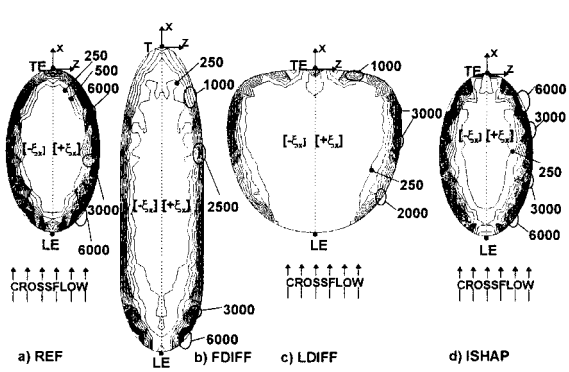

Fig. 10 The streamwise-aligned vorticity distribution at the film-hole exit plane shows the potential for generating secondary flow

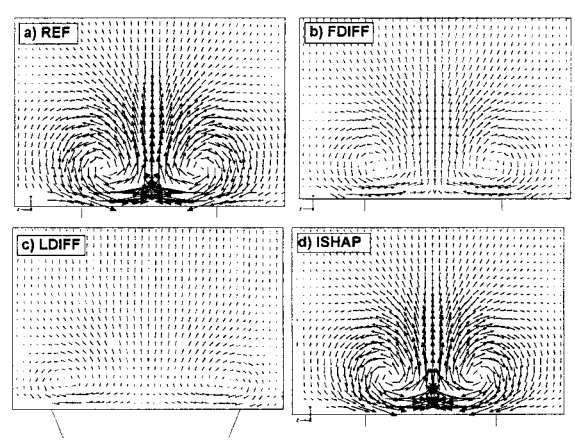

Fig. 11 A comparison of secondary flow magnitudes at $x/D = 2$ shows the near-elimination of streamwise vortices for the diffusion film holes

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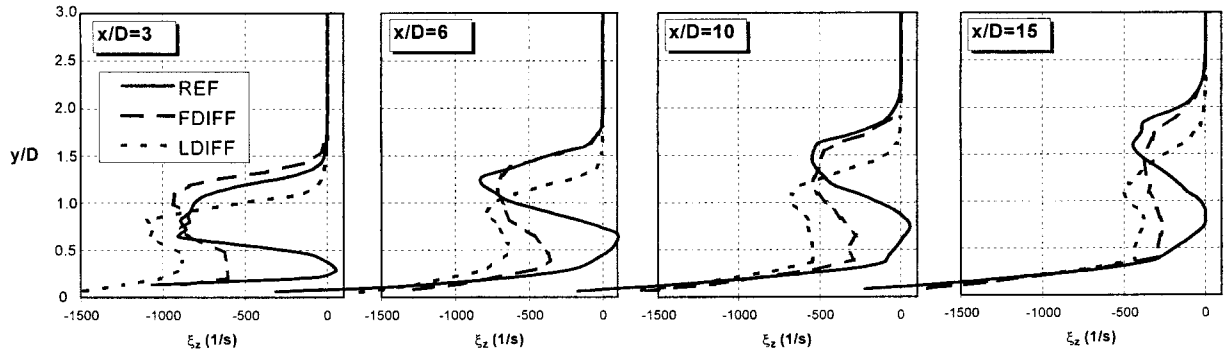

Fig. 12 Lateral (*z*) vorticity profiles delineate the film–crossflow shear layer and indicate the level of transverse velocity gradients for the REF, FDIFF, and LDIFF cases

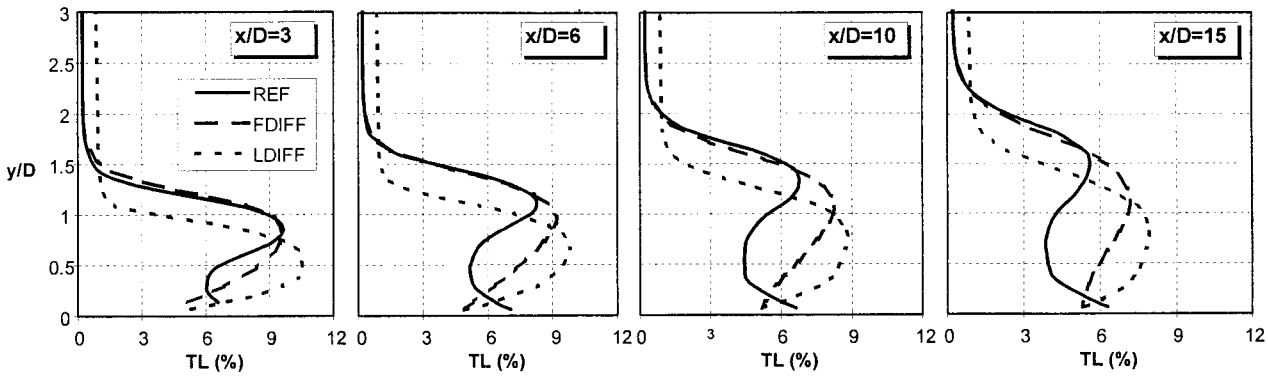

Fig. 13 Profiles of *TL* at downstream stations show the shear layer interaction between film and crossflow

contrast, the coolant jet in the REF case has already reached its lower limit by $x/D = 5$. The heat transfer coefficient for the LDIFF case is consistently higher (at all x/D) than the REF case for $M = 1.25$ and $M = 1.88$; significant differences are seen primarily in the near-field; as the jet moves downstream, only slight differences in the heat transfer coefficient are observed.

7.5 Inlet-Shaped Film Hole (ISHAP). In contrast to the previous shaped cases in which the inlet geometry is identical to the REF case, the ISHAP case demonstrates drastically different inlet flow behavior. The separation zone and the jetting region have been largely eliminated, which is the objective of this type of film-hole shaping. As a result, velocity gradients on the film-hole walls are relaxed slightly, and the throughflow velocity profile in the film hole is almost concentric. In addition, turbulence generation at the film-hole inlet is reduced to approximately two-thirds of the magnitude seen in the REF case; evidence of this is shown in Fig. 14.

Exit plane distributions of VR, α, and ξ_x (Fig. 6(*d*), Fig. 7(*d*), and Fig. 10(*d*), respectively) are very similar in appearance to the REF case, except for the elimination of the jetting region. Because of the elimination of high-momentum fluid on the windward side of the film hole, the crossflow is able to exert a considerable influence on the exit-plane velocity distribution. High-velocity fluid is forced to exit from the leeward side of the film hole, as the crossflow blocks the upstream side. This mechanism also contributes to intense positive and negative ξ_x at the leeward side of the film hole, in close proximity to the test surface. *TL* at the film-hole exit plane is approximately two-thirds that of the REF case, which denotes a much less viscous jet; this advantage is gained due to the elimination of turbulent production at the film hole inlet. Unfortunately, as discussed in the following paragraph, the lack of high *TL* at the film hole exit is inconsequential; for the range of *M* and *DR* studied in this work, the dominant turbulence source is due to the shear layer formed over the coolant jet.

The ISHAP case, by preventing the low-momentum/low-pressure zone at the inlet, eliminates much of the secondary motion

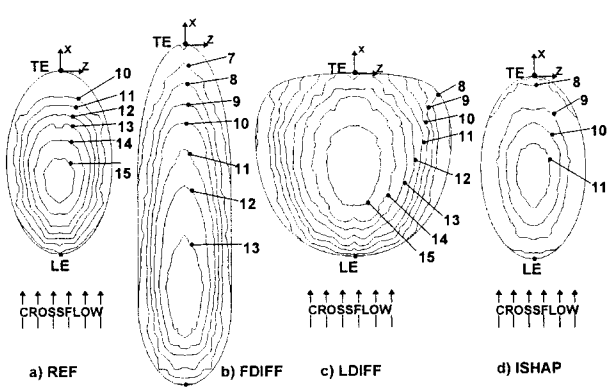

Fig. 14 A comparison of exit plane *TL* (percent) for all shapes shows the reduction of film hole turbulence by inlet shaping

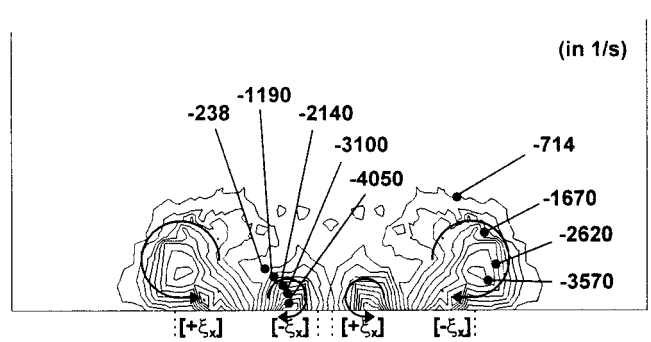

Fig. 15 Aligned vorticity contours at *x/D* = 0 for the CUSP case show the generation of an extra pair of vorticity cores at the cusp location

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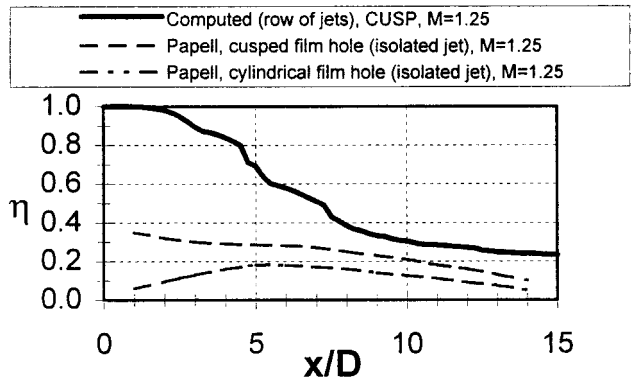

Fig. 16 A comparison of centerline effectiveness for computed (row of jets) and measured (single jet) results for the CUSP case

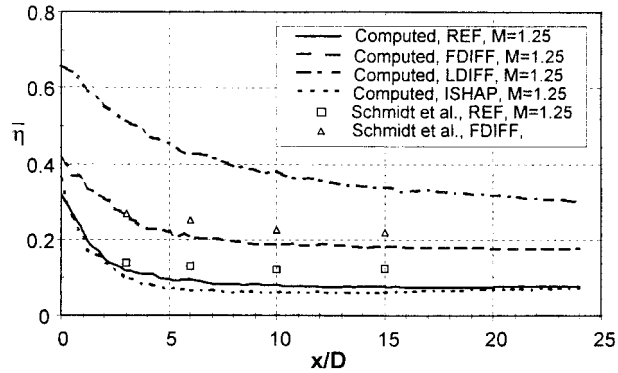

Fig. 18 A comparison between predicted and measured laterally averaged adiabatic effectiveness for various shaped hole film cooling configurations

within the film hole; the flow is purely in the throughflow direction. Since this secondary motion within the film hole creates negative streamwise-aligned vorticity (in a sense to cancel throughflow generated ξ_x; see Fig. 4(*a*)), no counteraction of the positive-aligned vorticity exists. So, the aligned vorticity advantage gained by relaxing velocity gradients near the sidewalls (by the elimination of the jetting region) is lost by this counteracting mechanism. Further details on this mechanism are given in section 8.2.

As such, the downstream behavior of the ISHAP case is almost identical to the REF case. Strong longitudinal vortices are formed, and the coolant jet lifts away from the surface. Due to less turbulent mixing in the coolant jet as it leaves the film hole, the cold temperature core is more intact in the near field; however, as the turbulence-laden film shear layer becomes the dominant mechanism, the coolant is dissipated just as in the REF case. Since the coolant has lifted away from the surface, this effect is not apparent in the surface temperature distribution.

The surface results for the ISHAP case strongly mimic the REF case, since the field in each case is very similar. For $M = 1.25$ and $M = 1.88$, the centerline and laterally averaged effectiveness match the REF case very closely. The same is true for the heat transfer coefficient; no significant differences are noted for the laterally averaged h for $M = 1.25$ or $M = 1.88$. So, the use of inlet shaping in this range of flow parameters is not useful for increasing downstream thermal protection.

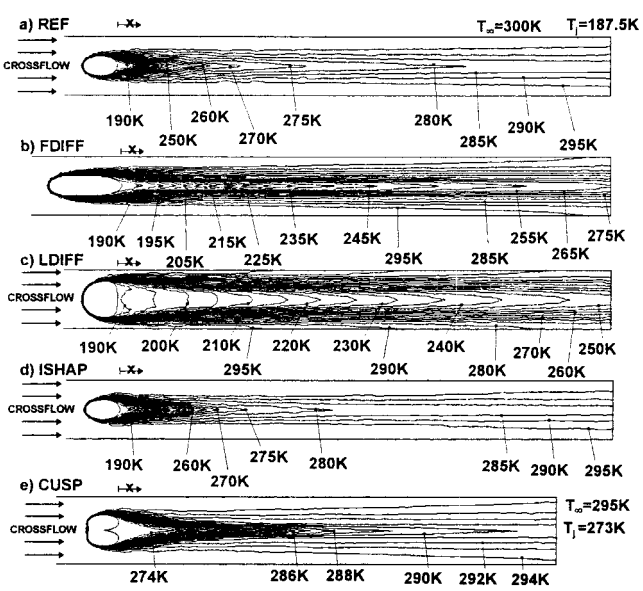

Fig. 17 A comparison of the temperature footprints on the test plate shows the distribution of coolant on the test surface for all film hole shapes

7.6 Cusp-Shaped Film Hole (CUSP). According to Papell's work, the direction of the film hole secondary motion is controlled by the orientation of the cusp-shaped film hole; this is outlined in section 2 (remember that the leeward-facing cusp-shaped hole, as shown in Fig. 1(*e*), is simulated in this work). No evidence to support Papell's conclusions concerning the direction of secondary motion within the film hole is found. Instead, it is believed that the direction of rotation is governed by the low-pressure/low-momentum zone at the entrance to the film hole near the downstream edge. Evidence from the simulations suggests that at this film hole inlet, boundary layer fluid along both the outer walls and the cusp wall is initially pumped downward toward the low-pressure zone. As the flow stabilizes downstream of the inlet, the secondary flow along the outer wall dominates that of the cusp wall; hence, the two chambers of the film hole each have a single vortex rotating in the direction away from the cusp. This direction is the same as that present in the REF case. Counterrotating structures are indeed present within the film hole chambers; however, their magnitude is approximately the same, when normalized by the throughflow, as the REF case.

At the exit plane of the CUSP film hole, the *VR* and *TL* distributions reflect the large L/D of this case. Due to the length of the film hole, inlet-generated turbulence is attenuated, and the only turbulence present at the exit plane is a weak "ring" generated by near-wall velocity gradients. Also, a thick boundary layer is present on all walls of the film hole, most notably on the straight windward section of the film hole.

Interesting vorticity characteristics are observed at the exit-plane of the CUSP film-hole. Along with the usual side-wall boundary layer vorticity observed in the other shaped film holes, the CUSP case also generates an extra pair of vorticity cores at the cusped exit, as shown in Fig. 15. This vorticity is contributed simply by the boundary layers that grow on the cusped wall. Unfortunately, this vorticity is destroyed quickly (by $x/D = 1$), since the two sides of the cusp are in close proximity; the two vorticity "pockets" emanating at each side of the cusp work to cancel one another. So, no influence is exerted on the longitudinal vortices formed as a result of the sidewall boundary layer vorticity.

Effectiveness results for the cusp-shaped film hole, shown in Fig. 16, do not agree well with data presented by Papell; however, computations are expected to overpredict the centerline effectiveness since the simulation was for a row of film-holes, whereas Papell's experimental data were taken for an isolated jet. Also note the centerline effectiveness magnitude of the reference film hole measured by Papell; although (1) *VR* is only slightly over unity and (2) $\alpha = 30$ deg is a very shallow injection angle, η shows an extremely degraded film even in the near field.

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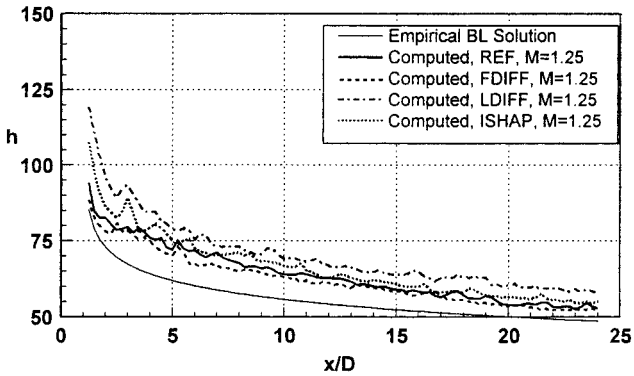

Fig. 19 Comparison of the laterally averaged heat transfer coefficient (W/m^2K) for various shaped hole configurations to an empirical correlation for a flat plate

8 Simulation Summary

8.1 Surface Results

8.1.1 Adiabatic Test Surface. The temperature footprint of each shaped hole configuration is shown in Fig. 17. It is evident that the REF and ISHAP cases are the worst performers in terms of effectiveness levels and lateral spreading, and that their behavior is almost identical. This implies, as discussed in section 7.5, that the alteration in the exit-plane distributions afforded by the ISHAP case does not have a significant impact on the film cooling performance. The FDIFF case performs very well along the centerline; however, lateral spreading characteristics of the jet, although better than the REF and ISHAP cases, are not favorable. Although the absolute values of lateral spreading are expected to be underpredicted due to the isotropic turbulence model used, strong internal consistency of the simulations enables reliable assessment of the relative performance between the different shaped cases. The LDIFF case is by far the best film cooling performer of the configurations tested in this work. The effectiveness is very high along the centerline, and lateral spreading and overall coverage of the test plate is impressive as the coolant jets appear to coalesce quickly as they move downstream. One would expect the coolant from this film hole to merge even more quickly than the simulations capture. The excellent performance of the LDIFF case is also due to its low trajectory and wide breadth; the coolant is effectively injected into the crossflow boundary layer and remains there until dissipated by diffusion mechanisms downstream. Also, no appreciable longitudinal vortices exist to lift the coolant core from the test surface. Moreover, the vortex cores are spread apart in the lateral direction, which further weakens induction lift of the coolant. The temperature footprint for the CUSP case in Fig. 17(*a*) shows the same characteristics as the REF case; the near field test section is cooled, but longitudinal vortices force liftoff of the jet/wake structure downstream.

A comparison of laterally averaged η is shown in Fig. 18; these effectiveness results reinforce the points given above, and show the relative performance of each film hole shape. Data for the REF and FDIFF cases also appear in Fig. 18; it is clear that the computations slightly underpredict the laterally averaged η by approximately 5 percent. This underprediction is due to the lack of lateral spreading captured by the computations, as documented in Walters and Leylek (1997). The injection widths of the REF and FDIFF cases are each only 1*D* wide; computations predict that the coolant is concentrated along the centerline at the expense of uncooled area between the coolant layers. Cooling performance (in terms of η) is in the same order for $M = 1.88$, but cannot be shown here due to space constraints.

8.1.2 Active Test Surface. The heat transfer coefficient for each film-hole shape is shown in Fig. 19 for $M = 1.25$. The thin line in this plot denotes the empirical unheated starting length boundary layer correlation for the heat transfer coefficient:

$$\mathrm{St} = 0.03\ \mathrm{Pr}^{-0.4}\ \mathrm{Re}_x^{-0.2}\left[1 - \left(\frac{x}{x_0}\right)^{0.9}\right]^{-1/9}$$

where St is the Stanton number, Pr is the Prandtl number, Re_x is the Reynolds number based on location, x is the distance from the start of the turbulent boundary layer, and x_0 is the unheated starting length. Each shape has a heat transfer coefficient somewhat above that of an undisturbed boundary layer. Notable differences in the h performance of the shaped film holes are only present in the near-field of the jet-crossflow interaction; downstream behavior is very similar for each case. Thus, film hole shaping tends to affect the h distribution in the immediate surroundings in the near field; far downstream, as the coolant jet travels as a wakelike structure, differences in h behavior are not significant.

For low and high M cases, the FDIFF case obtained the best (lowest) heat transfer coefficient, and the LDIFF case gave the highest h. For $M = 1.88$, all of the cases showed lower heat transfer coefficients than at $M = 1.25$. This result is simply because the crossflow velocity is lower for the high blowing ratio case. When normalized by the empirical solution above, however, the $M = 1.88$ cases exhibited slightly higher h on the test surface than the low M cases. Because the h results are very similar, the overall performance of the film cooling holes can be assessed by the adiabatic effectiveness distribution. In this regard, the LDIFF configuration has better cooling performance than the other configurations in this work; the centerline and laterally averaged effectiveness for this case is substantially higher than each other case.

8.2 Flowfield Results. For the diffusion-type film holes, the effective *VR* and α of the jet is lowered significantly (Figs. 6 and 7). A noticeable decrease in jet trajectory is not seen in the FDIFF case, however, since the strength of the jetting region at the exit plane is comparable to the magnitude seen in the REF case. Comparable effective *VR* is seen for the REF and ISHAP case; a notable difference in the ISHAP case is the elimination of the jetting region. This elimination allows the crossflow to exert considerably more influence on the jet exit velocity profiles; as shown in Fig. 6(*d*), the highest momentum jet fluid shifts from the windward side of the film hole to the leeward side.

TL is comparable between the REF and diffused cases; turbulence generated at the inlet plane of the film hole is the primary source of the turbulence observed at the exit-plane. The ISHAP case reduces film-hole turbulence a great deal, and might be useful in blowing parameter combinations in which film-hole turbulence dominates. For the range of *M* and *DR* studied in the present work, film-hole generated turbulence is inconsequential; mixing caused by the shear layers above the film layer is the dominant turbulent mechanism. In particular, cases in which the effective *VR* is lowered significantly (FDIFF and LDIFF cases) show that the coolant is dispersed by continuous turbulence generation in and throughout the film as it travels downstream in the far-field.

Clearly, the dominating mechanism determining downstream film cooling performance is the counterrotating vortex structure formed aft of the jet–crossflow interaction. The previous simulations demonstrate that the magnitude of this secondary motion is governed by the streamwise-aligned vorticity deposited from the film hole boundary layers. As such, one can conclude that the manipulation of the film-hole boundary layer shape factor and the direction of the outgoing boundary layer plays a crucial role in improving film cooling performance.

For a streamwise-injected jet, the contributors to ξ_x are illustrated in Fig. 4(*a*). Because of the secondary motion within the film hole, the interior boundary layer travels in a helical motion. The resultant vorticity vector, shown as the dotted line in Fig. 4(*a*), can therefore be decomposed into a component due to throughflow that is perpendicular to the flow direction, and a component due to

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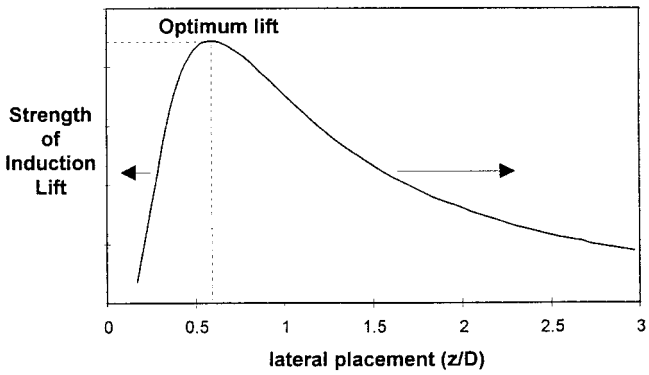

Fig. 20 A conceptual induction lift curve shows qualitatively the importance of avoiding coolant lift by the positioning of the deposited film hole boundary layer

secondary flow that is in the direction opposite to the flow direction. The resultant of these two components is then resolved to a coordinate system aligned with the predominant crossflow direction to determine ξ_x.

As evident in the simulations in this study, Fig. 4(*a*) illustrates two ways to decrease the aligned vorticity: (1) decrease the through-flow vorticity, or (2) increase the secondary vorticity. The former can be accomplished by relaxing the velocity gradient near the sidewalls of the film hole; this was illustrated in the FDIFF and LDIFF cases, and most diffusing film holes will have this characteristic. The latter can be accomplished by inducing strong secondary motion within the film hole. Papell (1984) claimed to have addressed this item with the windward cusp-shaped film hole; however, the present work was unable to substantiate his hypothesis regarding the mechanisms for any advantage gained by a cusp-shaped film hole (even though the case simulated was leeward-cusped). The idea of increasing the counterrotating motion within the film hole is promising; one could envision a case in which the secondary vorticity vector is strong enough to tilt the vorticity resultant so far as to create negative aligned vorticity at the film hole sidewalls. Such a condition would cause downstream counterrotating vortices, but in the *opposite* direction, such that the coolant is spread along the wall and negative induction lift is generated by the longitudinal vortices!

Figure 4(*b*) illustrates the effect on the aligned vorticity as the exiting flow angle decreases. The throughflow vorticity is tilted downward, where it cannot contribute a great deal to the aligned vorticity. This effect is illustrated most prominently by the FDIFF configuration. By the same argument, it is obvious that increasing the flow angle would not only increase the y-momentum and penetration into the crossflow, but would also tilt the resultant sidewall vorticity vector forward; this would (detrimentally) increase the magnitude of crossflow-aligned vorticity.

One might design a film hole configuration in which opposite-signed vorticity is generated in close proximity to the exiting sidewall boundary layers. In this case, vorticity cancellation would commence, and the aligned vorticity leading to the longitudinal vortices would be destroyed.

Last, the separation of the aligned vorticity deposited from the film hole is crucial. Induction lift generated by the longitudinal vortices can be minimized by positioning the exiting boundary layers an a suboptimal lift position. An example of this is illustrated in the LDIFF configuration. If opposite-signed vorticity is deposited in very close proximity, cancellation will occur; on the other hand, if the opposite-signed vorticity is deposited a large distance apart, induction lift is negligible. Therefore, as illustrated in Fig. 20, there must exist some "optimum lift" that must be avoided in film cooling configuration design.

9 Conclusions

A number of computational simulations are carried out for film cooling configurations involving cylindrical, forward diffused, laterally diffused, inlet shaped, and cusp shaped film holes. The key conclusions from this study are as follows:

- of the film holes tested in this work, the laterally diffused film hole provides the best coverage and highest effectiveness. The forward diffused film hole performs well along the centerline, but does not spread laterally. The cylindrical and inlet-shaped film hole perform poorly at both blowing ratios; the coolant lifts off from the test surface in these cases.
- film hole shaping can significantly affect the distribution of the exit plane variables, which determine downstream film cooling performance.
- crossflow-aligned vorticity from the film hole boundary layers, which governs the formation of detrimental longitudinal vortices, can be controlled by the geometry of the film hole. Streamwise vorticity reduction involves (1) decreasing the magnitude of the throughflow vorticity by decreasing velocity magnitude near the walls, (2) increasing the magnitude of the opposite direction vorticity by increasing the secondary motion within the film hole, or (3) changing the sense of the total vorticity vector by altering the exit flow direction.
- CFD can be used to sort the relative performance of shaped film cooling configurations and provide logic for determining the reasons for increases or decreases in film cooling performance.

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